greedy

MST => graph => spanning tree => MST

7-9 MST 90%

kruskals prims

35 hours

Lec_66

Prims Algorithm : it starts with nodes

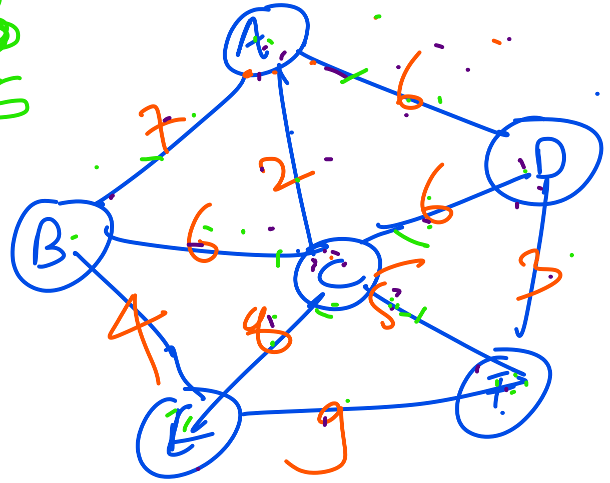
kruskals => start with edges

MST

$V = 6$

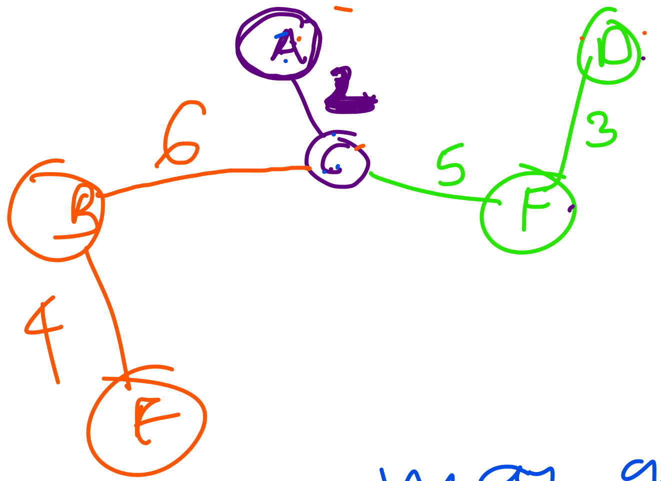
$E = 5$

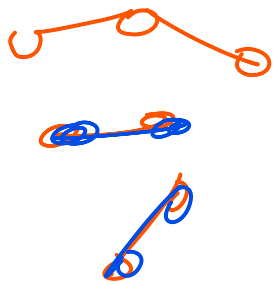
Ex1:



find MST and weight of MST using Prim's algorithm

Starts with a node





$= 4+6+2+5+3$

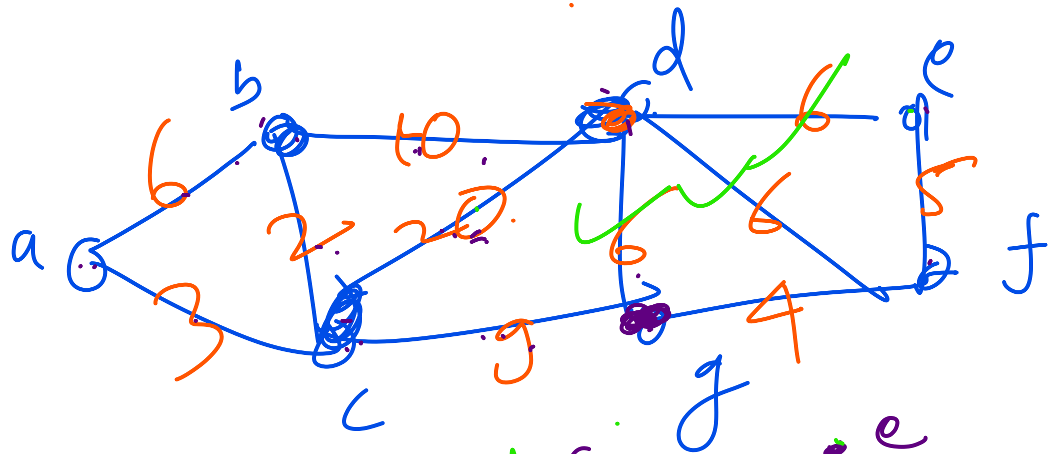
$= 20$

→ Kruskal algorithms in between may generate forest

→ Prim's always connected graph or tree

mean of

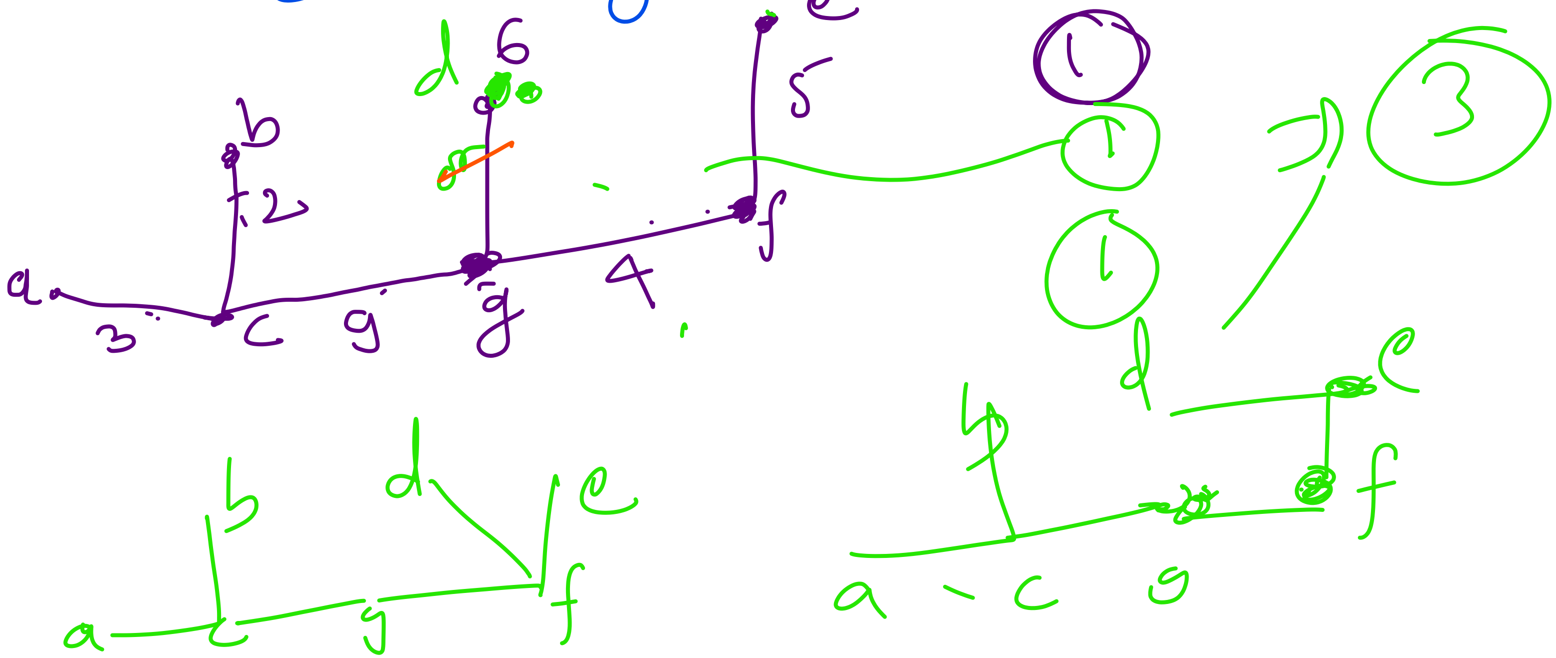
Ex2:



Start with any node

$V = 7$

$E = 6$



Algorithm and Time Complexity :

proof → this if difficult for those who are not aware of

Graph

adj list adj matrix

Representation graph

degree of graph

Time Complexity:

- Prims Algorithm is implemented by min heap data structure and graph is represented by adj. list
- In min heap each vertex contains 3 fields

node, min, ..

 1)Node

 2) min distance to neighbors

 3)All neighbors

- We know that in a min heap or max heap to insert ,delete ,increase or decrease the node ,it takes O(logn) in worst case
- TC= for creating min heap +Deleting in every Steps+ Checking degree of each node + decreasing degree

Tc= v+ vlogv +2E+2Elogv

$TC = V + V\log V + 2E + 2E\log V$

$V\log V + E\log V$

$\log V(V+E)$

$TC = (E+E)\log V$

$TC = 2E\log V = E\log V$

$TC = (V+E)\log V$

$E \geq V-1$

Proof of TC with example:

- Graph can be represented in 2 ways ,adj list or adj matrix
- Ex:

MST

TCX

Practical

Theory

Kahe gun offa
msg

→ Dijstra, Bellman, Warshal

Conclusion of TC:

\# which DS you are using

	Min heap	array	Sorted array
Adj list	(V+E)logv	V^2	EV
Adj matrix	V^2+Elogv	V^2	EV

representation of the graph

In Prim using adj list and array ...

If Adj list + Doubly linked list is used

- TC=O(VE)